PLATFORM OVERVIEW

Fuelize MDU

The on-vehicle unit that talks to the fuel dispenser. It takes an order's quantity, presets the dispenser, lets it pump exactly that amount, and reports the metered result back to the driver app.

What it is

Fuelize MDU runs on the Android tablet mounted at the fuel dispenser. It is the bridge between the Fuelize Driver App and the physical dispenser: when a delivery starts, the MDU receives the order, drives the dispenser to pump the right amount, and reports the measured result back.

It talks to the dispenser over a wired RS-485 serial connection. It supports two dispenser families - Tokheim (using the UDC protocol) and Eastman - and uses the right set of commands for whichever unit is connected.

It is an Android app written in Java. It keeps a small local record of every order it has dispensed, so the same order is not pumped twice, and shows a live screen of status, meter totals and the command exchange while fuel is flowing.

How it works

How the MDU takes an order and completes a fill.

RECEIVE ORDER

An order arrives

1. The Driver App opens the MDU with the order details - quantity, order number and a verification code.
2. The MDU checks it has not already dispensed this order.
3. It connects to the dispenser and begins reading its status.

VERIFY

Confirm before fuelling

1. The order is verified by OTP, or by RFID tag where a reader is fitted.
2. The MDU reads the dispenser's unit price.
3. It shows the preset amount and waits for the nozzle to be lifted.

DISPENSE

Preset and pump

1. The MDU sends the ordered quantity to the dispenser as a preset and authorises the fill.
2. The dispenser pumps and stops automatically at the ordered amount.
3. The live screen shows the quantity as it climbs.

CLOSE

Record and report back

1. The MDU reads the start and end meter totals and works out the delivered quantity.
2. It saves the order locally as dispensed.
3. It sends the dispensed quantity and meter totals back to the Driver App.

What it covers

The MDU is made up of the functional parts below.

Part	Modules	What you can do
Dispenser control	Tokheim (UDC) driver, Eastman driver	Speak each dispenser's command set over RS-485: read status and price, preset and authorise, start, pause or resume, stop and reset.
Order handling	Order intake, OTP / RFID verify	Receive an order from the Driver App, verify it by OTP or RFID tag, and prevent the same order being dispensed twice.
Metering	Preset, Auto-stop, Totalizer read	Set the exact quantity, stop automatically when it is reached, and read the start and end meter totals to confirm what was delivered.
Reporting	Result callback, Live status & log	Send the dispensed quantity back to the Driver App and show a live screen of status, totals and the raw command exchange.

Part	Modules	What you can do
Device	Local record, Serial port, Kiosk mode	Keep a local log of orders dispensed on this unit, manage the RS-485 serial connection, and run locked to the app on the tablet.

How it runs

Layer	Detail
Application	An Android app written in Java, running on the tablet mounted at the dispenser.
Dispenser link	Wired RS-485 serial - Tokheim at 9600 baud (UDC, with Doubletalk encoding) and Eastman at 2400 baud.
On-device data	A local database on the tablet (SQLite) holding the orders dispensed on this unit.
Order link	Launched by the Driver App through a device link that carries the order, and returns the dispensed amount the same way.
Verification	Order confirmed by OTP, or by RFID tag where a reader is fitted.